

Chronic & Rare-Illness Patient Advocacy Tool

Status: Draft for Transition into UX for Healthcare

Owner: Lesly

Version: v1 (prototype scope only)

1. Problem Statement

Patients with rare, complex, or hard-to-diagnose chronic conditions struggle to be heard by doctors during appointments. Diagnosis-by-exclusion pathways and inconsistent treatment gatekeeping mean patients often arrive at appointments with fragmented, hard-to-communicate symptom histories; and leave without being believed or without a clear next step.

Evidence problem: patients don't have a structured, credible way to show a doctor what's actually happening to their body over time.

2. Primary User

A person managing an undiagnosed or hard-to-diagnose chronic condition who has already had at least one dismissive or inconclusive doctor visit. They are not looking for a diagnosis from the app; they're looking for a way to walk into the next appointment with evidence a doctor can't easily wave away.

3. Evidence Base → Design Implications

Research signal	What it tells you	Design implication
"Rare" / diagnosis-by-exclusion comments	Patients are told their case doesn't fit standard patterns, which stalls belief and action	The tool needs to produce something that looks like <i>clinical evidence</i> , not a diary — format matters as much as content
Treatment-gatekeeping contradiction	Patients get inconsistent reasoning for why treatment is or isn't offered	Output should make patterns visible enough that inconsistency in a doctor's reasoning becomes harder to justify — this is about leverage, not persuasion

Food-cycling comment	Symptoms don't move linearly; they cycle, relapse, and correlate with triggers over days/weeks	Tracking must capture cycles and correlations, not just daily severity scores. A flat log misses the exact pattern doctors need to see.
Community post	Undirected self-research produces exhaustive but undirected findings – no way to prioritize which hypothesis to chase	v1 should NOT be an open research/community tool. That's the failure mode to design against, not toward

4. V1 Scope

In scope

- Structured symptom + trigger logging (not just severity; captures what precedes/follows a flare)
- Pattern view that surfaces cycles and correlations over time (this is the food-cycling insight made visible)
- One-tap export/summary formatted for a doctor visit – the "evidence artifact"
- Basic tagging for suspected triggers (food, sleep, stress, medication, environment) so correlations can surface without the user having to run their own analysis
- AI-assisted correlation detection (pattern view) and AI-drafted narrative (doctor report) – both user-toggable off at any time (see section 9)

5. Core User Flows (v1)

1. Log a symptom event – quick entry, low cognitive load (chronic illness users are often fatigued/brain-fogged when logging)
2. Log a possible trigger – food, sleep, stress, medication, environment – tied to timeframe, not just same-day.
3. View pattern over time – cycles, correlations, flare frequency – this is where the food-cycling insight lives.
4. Generate doctor-ready summary – the artifact the user brings to the appointment
5. (v2) Post-appointment note – what the doctor said/decided, so the next cycle of tracking has context

6. Assumptions to Validate

Ranked by how much they'd sink the concept if wrong:

1. Patients want a structured artifact for the doctor, not just personal tracking. (If false, this is a wellness app, not an advocacy tool, a different product entirely.)
2. Cyclical/correlation tracking is something patients will actually do consistently, not just log severity once and abandon.
3. A doctor-facing summary changes how the appointment goes. (Hardest to test as a student project — even one real patient's account of "I brought data to a doctor and X happened" would be gold.)

7. AI-Powered Feature

Where AI is used:

- Pattern view: AI-assisted correlation detection across logged symptoms and triggers — surfaces relationships a manual review might miss (e.g. "flares cluster 2–3 days after dairy" style patterns), building directly on the food-cycling insight from research.
- Doctor report: AI drafts a plain-language narrative summary from the structured log data — turning raw entries into the kind of coherent account a doctor can skim in the time of an appointment.

Control: Both are user-toggleable, visibly off or on; not buried in settings. AI assistance can be turned off at any time; when off, the pattern view falls back to a plain chart/table and the doctor report falls back to a raw structured log.

This matters for two reasons: (1) trust; some users won't want an algorithm interpreting their health data, and that's a legitimate choice; (2) credibility; a doctor is more likely to take a summary seriously if the patient can say plainly whether AI wrote part of it.

Design implication: any AI-generated content (correlation call-outs, summary narrative) needs a visible, unambiguous marker that it's AI-generated; not blended in as if it were raw data. This is a trust requirement, not a legal footnote.

8. Accessibility Requirements

This audience is often navigating fatigue, brain fog, pain, or a bad day when they open this app; accessibility here is core to whether the tool works for the people it's for.

- Touch targets: minimum 44×44pt (iOS) / 48×48dp (Android) for all interactive elements — bigger where possible, since fine motor control can be affected by pain or fatigue.
- Color contrast: WCAG 2.2 AA as the floor (4.5:1 body text, 3:1 for UI components and large text) on every screen. AAA (7:1) is worth aiming for on body text and key labels, but holding the *whole* UI to AAA isn't realistic; data visualizations especially can't always meet it. (to check)
- Colorblind-safe: never encode meaning by color alone, especially in the pattern view; pair color with a second cue (icon, pattern/hatching, direct label). This applies to flare severity, trigger categories, and any "better/worse" trend indicator.
- Plain language: aim for a reading level that doesn't require medical literacy. Avoid clinical jargon in the UI even though the output is doctor-facing; the *input* experience is for the patient.
- Low cognitive load: one clear action per screen, no dense forms, no more than one decision at a time. Direct extension of Craft Checkpoint C1; if you can remove an element and the screen still works, remove it.
- Tone: straightforward, calm, never alarming or gamified. This is someone who may already be scared or exhausted; no urgency-driven UI patterns (streaks, red warning banners for missed logs, guilt-driven notifications).

9. Questions for Feedback

1. Is MCAS specific enough as a v1 archetype, or should I pick a condition with more publicly-documented patient/doctor friction to design against (e.g. POTS, endometriosis, fibromyalgia)?
2. What does "doctor-ready" actually mean in a clinical context — is there a real format (like a SOAP note structure) I should be designing toward, or is a clear narrative + data summary enough?
3. Does deferring the community/peer-support pillar out of v1 read as a reasonable scoping call; or is comfort from others' shared experiences too core to patient advocacy to leave out entirely, even if the "undirected self-research" failure mode is handled some other way?
4. How should the pattern view actually show cycles and correlations: a chart, a calendar-style heatmap, or something else; that stays readable for someone who's fatigued or brain-fogged?

10. Figma Screens

<https://www.figma.com/design/avRzFZZb8urgnwYBpVc17s/HealthcareApp?node-id=41-230>